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FoodEx2 maintenance 2016-2018

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Abstract

FoodEx2 is a comprehensive food classification and description system released by EFSA in 2011 (revision 1) and replaced in April 2015 by FoodEx2 revision 2, as a consequence of a testing phase. A first maintenance of the system was carried out in 2015 and a second during the years 2016-2018 in order to evaluate further comments and proposals provided to EFSA by users and stakeholders. This technical report describes the outcome of the second maintenance process, including addition of new terms, disposal of existing terms and amendments to existing terms including enrichment of implicit facets.

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Summary

A standardised food classification and description system is a key element to provide a common link to data sources coming from different food safety domains. In order to meet this requirement, in 2011 FoodEx2 (revision 1) was published by EFSA followed by a major revision in 2015 (FoodEx2 revision 2). Although the second revision achieved a good level of stability, innovation in the food market and accumulated experience by users led to continuous need for updates in the system.

A maintenance technical group composed of EFSA staff with expertise in different domains (food consumption, chemical contaminants, pesticides residues, etc.) was set-up to evaluate proposals and requests provided by users and stakeholders during the period 2016 – 2018 and take decisions regarding its implementation.

Changes in the terminology and the structure of the system were applied. 680 new terms were added in different hierarchies and facets, the scope of some existing terms was better specified or expanded by changing the name, scientific name or adding alias, 36 terms were deprecated due to duplication and 38 terms were identified as redundant and were dismissed or set as not reportable. The applicability of some terms was extended to further hierarchies and/or facets and the position of some terms in the hierarchy tree was changed since a better parent-child term relationship was found. Furthermore, the structure and content of some the FoodEx2 sections was amended, insects were refined and extended with new terms, some terms related to the varieties of *Musa* spp. (banana) were revised in order to make a better distinction between different species, the catalogue was revised and updated in line with the updated Annex I of Regulation (EC) No 396/2005 and matrix codes were updated for 50 terms, the botanicals hierarchy was significantly expanded in order to harmonise the catalogue according to the latest version of the EFSA's Compendium of Botanicals and the section containing Bird species was expanded to enable data collection on Avian Influenza.

Additional activities on the FoodEx2 catalogue were carried out. A mapping between FoodEx2 codes and FoodEx codes was performed in order to harmonise previously reported occurrence and consumption data in FoodEx. FoodEx codes were implemented in the browser as implicit attribute 'foodexOldCode'. A major revision of the catalogue was performed in the area of implicit facets; all implicit facets were checked and updated in the system and in the Data Collection Framework (DCF). A complete revision of the "scope notes" was implemented with the purpose of improving the information provided and including external links. LanguaL codes, previously attributed to the FoodEx2 terms, were reviewed and if necessary, updated and new LanguaL codes were assigned as implicit attributes to the FoodEx2 facet terms. Finally, the FoodEx2 catalogue was extended with a new hierarchy FeedAddExpo. This hierarchy is neither reportable nor available for usage in any data collection but was created to support the development of the Feed Additives Consumer Exposure (FACE) calculator used for the estimation of dietary exposure to residues of feed additives (and their metabolites) present in food of animal origin.

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1. Introduction

1.1. Background and Terms of Reference as provided by the requestor

In 2011, EFSA released a comprehensive Food Classification and Description System (named FoodEx2 revision1) (EFSA, 2011a, b). This system was developed with the aim of harmonising the description of the food matrices within data collections in different food safety domains. Following the publication, a testing phase was carried out in order to highlight strengths and weaknesses of the classification and to identify possible issues and needs for refinement. Based on the outcome of the testing phase, EFSA revised the system and in 2015 the second revision of the food classification and description system (FoodEx2 revision2) was published (EFSA, 2015).

According to the guidance documents on FoodEx2 (EFSA, 2011a, 2015), the users should provide to EFSA proposals for implementation and maintenance in order to refine the system and an annual EFSA maintenance working group should be set up in order to evaluate the suggestions and decide on their implementation. In fact, innovation in the food market and gained experience by using the system lead to the need of keeping the catalogue updated.

By using the system, Member States and other stakeholders, including internal users in EFSA, provided constructive comments and requests for addition of missing terms. These suggestions were taken into account for the maintenance purposes. A technical group composed of EFSA staff with expertise in different domains carried out the maintenance process evaluating whether and how to implement the received suggestions in the FoodEx2 system.

A first maintenance has already taken place in 2015 (EFSA et al., 2016a). The present report describes the outcome of the second maintenance process in order to keep track of the changes in the system.

2. Data and Methodologies

2.1. Data

The FoodEx2 catalogue version published in May 2016 was used as starting point. The suggestions provided during the period between the beginning of 2016 and end of 2018, by the users and stakeholders of the FoodEx2 system, were collected through written feedback and used to perform the maintenance.

2.2. Methodologies

Each request for addition or change in the structure was considered and, on the basis of expert judgment, the maintenance group decided whether a change was needed or whether the request was already covered by the available terms and facet descriptors. All the changes were made following the structure and the logic of the system as described in the report on the development of FoodEx2 revision 2 (EFSA, 2015).

2.2.1. Tools

The tools used for implementing the maintenance were Microsoft Excel® and the FoodEx2 Catalogue Browser¹ (EFSA, 2018), a user friendly tool developed in Java by staff of EFSA (Evidence Management (DATA) Unit) to help creating FoodEx2 codes and to facilitate the management of the catalogue.

2.2.2. Maintenance process

In order to carry out this second maintenance, the maintenance group performed the listed steps:

- Evaluating comments and proposals

¹ <https://www.efsa.europa.eu/en/data/data-standardisation>

- Deciding on the implementation of the proposals based on the following options: adding new terms, extending or better specifying the scope of existing terms or identifying the correct way of implementing the suggestion using already existing terms and rules, deprecating terms and set terms as dismissed or not reportable.
- Checking for inconsistencies and ambiguities
- Updating the FoodEx2 catalogue
- Summarising the changes for the present technical report.

3. Maintenance output

3.1. Addition of new terms

The terms added in the 2016-2018 maintenance were in total 680. Areas that gained significant number of terms were related to the hierarchy of botanicals, which gained 314 new terms (see section 3.6), Avian influenza data collection (see section 3.7) and insects (see section 3.3).

Minor additions were also performed in different hierarchies and facets related to the following areas:

- Transmissible Spongiform Encephalopathy (TSE)/ Bovine Spongiform Encephalopathy (BSE) data collection
- Chemical hazards database
- Pesticides residues data collection
- Varieties of plastic used as a packaging materials
- Molecular typing data collection
- Exposure data to pesticides for residents and bystanders and environmental risk assessment
- Food additives data collections.

A detailed table with all additions performed during 2016-2018, the hierarchies and facets affected and the motivation behind is available in Appendix A.

3.2. Changes in existing terms

Taking into account all suggestions from users and keeping the consistency of the system, amendments to existing terms were applied. Term extended names, common names and/or scientific names were changed for approximately 10.000 terms. These changes were mainly editorial changes, such as singular/plural or correction of misspelling, addition of new common names/aliases and removal of existing common names/aliases. These minor changes are not presented in this report, but they were implemented, therefore they are included in the FoodEx2 database and appear in the file 'MTX_10.0.xlsx' spreadsheet 'term' containing the complete content of the FoodEx2 system available as Appendix B.

For 54 existing terms the extended term name was changed to better specify the scope of the term or to expand its meaning (Table 1). However, in some cases the old name was kept as alias while a new one was added as the term extended name.

Table 1: Changes performed on FoodEx2 term extended name in the 2016-2018 maintenance in order to better specify the scope of the term or to expand its meaning

Code	Old term extended name	New term extended name
A003K	Rye flour, light	Rye flour, refined
A004J	Spelt flour, light	Spelt flour, refined
A009G	Choux pastry doughs	Choux pastry dough and similar
A00KX	Lettuces	Lettuces (generic)
A00XQ	Lemon myrtle	Lemon myrtle (fresh herb)

Code	Old term extended name	New term extended name
A00ZJ	Pickled/marinated vegetables	Pickled / marinated vegetables
A016D	Other oilseeds (including elsewhere non-listed seeds)	Other seeds (including elsewhere non-listed oilseeds and other seeds)
A019X	Cassia bark	Cassia bark (generic)
A01FE	Bilberries	Bilberries (generic)
A01FF	Bilberries	Bilberries (European blueberries)
A01ZK	Animal other organs (edible offal non-muscle)	Animal edible offal, non-muscle, other than liver and kidney
A01ZM	Bovine edible offal (other than liver and kidney)	Bovine edible offal, non-muscle, other than liver and kidney
A01ZN	Buffalo, other organs (edible offal non-muscle)	Buffalo edible offal, non-muscle, other than liver and kidney
A01ZP	Pig edible offal (other than liver and kidney)	Pig edible offal, non-muscle, other than liver and kidney
A01ZQ	Sheep edible offal (other than liver and kidney)	Sheep edible offal, non-muscle, other than liver and kidney
A01ZR	Goat edible offal (other than liver and kidney)	Goat edible offal, non-muscle, other than liver and kidney
A01ZS	Equine edible offal (other than liver and kidney)	Equine edible offal, non-muscle, other than liver and kidney
A01ZT	Horse, other organs (edible offal non-muscle)	Horse edible offal, non-muscle, other than liver and kidney
A01ZV	Asses-mules-hinnies, other organs (edible offal non-muscle)	Asses-mules-hinnies edible offal, non-muscle, other than liver and kidney
A01ZX	Rabbit, other organs (edible offal non-muscle)	Rabbit edible offal, non-muscle, other than liver and kidney
A01ZZ	Camel, other organs (edible offal non-muscle)	Camel edible offal, non-muscle, other than liver and kidney
A020B	Deer, other organs (edible offal non-muscle)	Deer edible offal, non-muscle, other than liver and kidney
A020C	Wild boar, other organs (edible offal non-muscle)	Wild boar edible offal, non-muscle, other than liver and kidney
A020E	Poultry edible offal (other than liver and kidney)	Poultry edible offal, non-muscle, other than liver and kidney
A020F	Chicken, other organs (edible offal non-muscle)	Chicken edible offal, non-muscle, other than liver and kidney
A020G	Turkey, other organs (edible offal non-muscle)	Turkey edible offal, non-muscle, other than liver and kidney
A020H	Duck, other organs (edible offal non-muscle)	Duck edible offal, non-muscle, other than liver and kidney
A020J	Goose, other organs (edible offal non-muscle)	Goose edible offal, non-muscle, other than liver and kidney
A020K	Ratites other organs (edible offal non-muscle)	Ratites edible offal, non-muscle, other than liver and kidney
A027G	Carp, indian	Carp, indian (generic)
A028A	Tilapia	Tilapias and similar
A02BJ	Anglerfish and monkfish	Anglerfish, monkfish and stargazers
A02KR	Frogs (legs)	Frogs meat
A03ZR	Pizza and similar with fish topping	Pizza and similar with fish/seafood topping
A040N	Pasta based dishes (cooked)	Pasta based dishes, cooked
A044C	Tomato-containing cooking sauces	Tomato-containing cooked sauces
A066M	Plant commodities-related (as part-nature)	Plant commodities-related food (as part-nature)
A069C	Animal commodities-related (as part-nature)	Animal commodities-related food (as part-nature)
A069L	offal and other slaughtering products (as part-nature)	Offal and other slaughtering products (as part-nature)
A06AV	Preserved sausages (as part-nature)	Preserved or partly preserved sausages (as part-

Code	Old term extended name	New term extended name
		nature)
A0C2K	FA-12.3 Vinegars	FA-12.3 Vinegars and diluted acetic acid (diluted with water to 4-30 % by volume)
A0CVC	Other frog legs from frogs not belonging to the genus <i>Rana</i>	Other frog meat from frogs not belonging to the genus <i>Rana</i>
A0CVE	Frog legs (<i>Rana</i> genus)	Frog meat (<i>Rana</i> genus)
A0DKN	Baby leaf Chards	Baby leaf chards
A0ESH	Sand (as part-nature)	Sand-related (as part-nature)
A0ESK	Pollen (as part-nature)	Pollen-related (as part-nature)
A0ESS	Leaves (tea and other) used for infusions or hot drinks	Leaves used for infusions or hot drinks
A0ETS	Candied or sugar-preserved vegetables	Candied or sugar preserved vegetables
A0EZK	Fried or extruded cereal or root-based products	Fried or extruded cereal, seed or root-based products
A0F0Y	Feed (as part-nature)	Feed-related (as part-nature)
A0F3L	Mammals other organs (edible offal non-muscle)	Mammals edible offal, non-muscle, other than liver and kidney
A15NE	<i>Secale cornutum</i> (live plant/fungus)	<i>Secale cornutum</i> (as plant/fungus)
A16EQ	Puddings based on cereal products or bread	Puddings based on cereal products and/or nuts/fruits and similar
A16GF	Rice, fish and vegetable based dishes	Rice, fish/seafood and vegetable based dishes

Few general changes were also applied such as: the description 'live organisms' is replaced by the description 'as organism', 'live animals' is replaced by 'as animals', 'live plants' is replaced by 'as plant' and for some feed items the extension (feed) was added. The type of term² was changed for 50 terms related to the pulses to enable correct applicability of the matrix codes in line with the Annex I of the Regulation (EC) No 396/2005 (section 3.5). These minor changes are not presented in this report, but they were implemented, therefore they are included in the FoodEx2 database and appear in the file 'MTX_10.0.xlsx' spreadsheet 'term' containing the complete FoodEx2 catalogue available as Appendix B.

3.3. Amendments on insects and arachnids sections

National food composition tables in South East Asia and some in African countries include insects. Moreover, the food composition unit at Food and Agriculture Organisation (FAO) of the United Nations in Rome, INFOODS, has compiled a list of nutrient composition of indigenous foods in various countries, which includes a number of insects. In order to provide a common link to all the diverse food databases FoodEx2 was populated with the following new terms as children of the previously general term Insects [A02LQ]: Edible insect eggs [A16XB], Edible insect larvae [A16XC], Edible insect pupae/nymphs [A16XD] and Edible insect adults [A16XE]. In addition, the term Silkworm [A16QJ] that includes edible parts of Silkworm, classified under the species *Bombyx mori*, was introduced under Edible insect larvae. Finally, a separate code was generated for Arachnids [A16XF] at the same level as Insects. All above mentioned terms were introduced in the MTX (FoodEx2 Matrix), Reporting, Zoonoses, Exposure and VetDrugRes hierarchies, facet F27 - Source-commodities and F04 - Ingredient and were set as raw primary commodities.

Certain insect species were also added under the facet F01 - Source. Some of these insect species were mentioned in the EFSA's opinion on the risk profile related to production and consumption of insects as food and feed (EFSA Scientific Committee, 2015). The remaining species were selected in collaboration with FAO. All terms added under this section are detailed in Table 2.

Table 2: New FoodEx2 terms related to insects and arachnids

Code	Name	Reportability - Hierarchies/Facets
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² The type of the term can be one of the following: Natural sources, Raw Primary Commodities (RPC), RPC Derivatives, ingredients, Composite food simple, Composite food aggregated, Broad or mixed, Facets

Code	Name	Reportability - Hierarchies/Facets
A16QJ	Silkworm	MTX (FoodEx2 Matrix), Reporting, Zoonoses, Exposure, VetDrugRes/ Source-commodity (F27), Ingredient (F04)
A16XB	Edible insect eggs	MTX (FoodEx2 Matrix), Reporting, Zoonoses, Exposure, VetDrugRes/ Source-commodity (F27), Ingredient (F04)
A16XC	Edible insect larvae	MTX (FoodEx2 Matrix), Reporting, Zoonoses, Exposure, VetDrugRes/ Source-commodity (F27), Ingredient (F04)
A16XD	Edible insect pupae/nymphs	MTX (FoodEx2 Matrix), Reporting, Zoonoses, Exposure, VetDrugRes/ Source-commodity (F27), Ingredient (F04)
A16XE	Edible insect adults	MTX (FoodEx2 Matrix), Reporting, Zoonoses, Exposure, VetDrugRes/ Source-commodity (F27), Ingredient (F04)
A16XF	Arachnids	MTX (FoodEx2 Matrix), Reporting, Zoonoses, Exposure, VetDrugRes/ Source-commodity (F27), Ingredient (F04)
A16RJ	Age classes for insects	MTX (FoodEx2 Matrix)/ Animal-age-class (F31)
A16RK	Insect eggs	MTX (FoodEx2 Matrix)/ Animal-age-class (F31)
A16RL	Insect pupae/nymphs	MTX (FoodEx2 Matrix)/ Animal-age-class (F31)
A16RM	Insect larvae	MTX (FoodEx2 Matrix)/ Animal-age-class (F31)
A16RN	Insect adults	MTX (FoodEx2 Matrix)/ Animal-age-class (F31)
A16SG	Blattodea (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16SH	Termites (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16SJ	Jewel beetles (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16SK	Palm Weevil (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16SL	Diving beetles (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16SM	Christmas Beetle (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16SN	Copris dung beetles (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16SP	Large dung beetles (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16SQ	Scarab beetles (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16SR	Dung beetles (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16SS	June Bugs (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16ST	Lesser mealworm (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16SV	Mealworm beetle (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16SX	Superworm beetle (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16SY	Giant mealworm beetle (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16SZ	Blow fly (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16TA	House fly (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16TB	Black soldier fly (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16TC	Aphids (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16TD	Giant Water Bug (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16TE	Cicadas (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16TF	Jumiles (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16TG	Bumblebees (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16TH	Leafcutter ants (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16TJ	Honey pot ant (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16TK	Lemon ant (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16TL	Hornets (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16TM	Lepidoptera (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16TN	Agave worm (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16TP	Cossid moth (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16TQ	Bamboo Caterpillar (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)

Code	Name	Reportability - Hierarchies/Facets
A16TR	Agave Snout Weevil (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16TS	Maguey worm (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16TT	Bagworms (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16TV	Lesser wax moth (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16TX	Greater wax moth (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16TY	Indian meal moth (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16TZ	Mopane worm (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16VA	Emperor moths (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16VB	Tomato hornworm (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16VC	Hornworm (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16VD	Dragonflies (odonata) (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16VE	Orthoptera (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16VF	House cricket (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16VG	Short tail cricket (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16VH	Banded cricket (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16VJ	Field cricket (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16VK	Black cricket (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16VL	Teleogryllus crickets (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16VM	Acrida locusts (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16VN	African rice grasshoppers (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16VP	Migratory locust (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16VQ	Spur-throat grasshoppers (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16VR	Oxya grasshoppers (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16VS	American grasshopper (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16VT	Chapulines (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16VV	Bush crickets (as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16VX	Stoneflies (Plecoptera)(as animal)	MTX (FoodEx2 Matrix)/ Source (F01)
A16VY	Caddisflies (Trichoptera)(as animal)	MTX (FoodEx2 Matrix)/ Source (F01)

Moreover, five new facet descriptors related to insects are introduced in the MTX (FoodEx2 Matrix) hierarchy and facet F31 - Animal-age-class (See Appendix A for detailed list).

3.4. Amendments on terms related to the varieties of *Musa* spp. (banana)

The structure of the FoodEx2 section related to the raw primary commodities obtained by plants of the species *Musa* spp. was revised. The main criteria taken into account in the revision process were the edible varieties of *Musa* spp. cultivated in different areas of the world and the characteristics of the fruits. The edible cultivars available on the market are hybrids and polyploids of two species, *Musa acuminata* and *Musa balbisiana*. The most cultivated species are hybrids and are called *Musa x paradisiaca*. Fruits from some cultivars are high in sugar and usually eaten raw as fresh fruit, while fruits from other cultivars are high in starch and, though edible raw when very ripe, they are normally consumed as cooking staple food in many tropical countries. The former, known as common bananas or dessert bananas, are the fruits obtained from non-hybrid cultivars of *Musa acuminata* (*Musa acuminata* AA, AAA, AAAA groups) and from cultivars of hybrid species (*Musa x paradisiaca* AB, AAB, ABB, ABBB, AAAB, AABB groups). The latter, known as plantains or cooking bananas, are mainly obtained from triploids hybrids (*Musa x paradisiaca* AAB, ABB groups), and in few cases from non-hybrid cultivars of *Musa balbisiana* (BB and BBB groups) and *Musa Acuminata* (AAA group). All changes related to this section are presented in Table 3.

Table 3: Changes performed to the section related to the varieties of Musa spp. (banana) section

FoodEx2 code	Term extended name (old)	Scientific name (old)	Common name (old)	Term extended name (new)	Scientific names (new)	Common names (new)
A064N	Banana AB hybrid cultivars (live plants)	Musa x paradisiaca L.	Plantains	Banana AB hybrid cultivars (as plant)	Musa x paradisiaca L.\$Musa acuminata x Musa balbisiana	Plantains
A0DEJ	Banana flowers (x paradisiaca)	hybrids Musa acuminata x Musa balbisiana		Banana flowers - paradisiaca cultivars	hybrids Musa acuminata x Musa balbisiana	
A0DEK	Banana flowers (balbisiana)	Musa balbisiana Colla		Banana flowers - balbisiana cultivars	Musa balbisiana Colla	
A0DEL	Banana flowers (acuminata)	Musa acuminata Colla		Banana flowers - acuminata cultivars	Musa acuminata Colla	
A0DQG	Plantains paradisiaca	Musa acuminata x Musa balbisiana		Plantains - paradisiaca cultivars	Musa acuminata x Musa balbisiana\$Musa x paradisiaca	Plantains paradisiaca\$African Plantains\$Saba banana
A0DQH	Plantains balbisiana	Musa balbisiana Colla		Plantains - balbisiana cultivars	Musa balbisiana Colla	Plantains balbisiana
A0DQJ	Plantains acuminata	Musa acuminata Colla		Plantains - acuminata cultivars	Musa acuminata Colla	Plantains acuminata\$east African highlands banana
A0DQK	Cavendishes paradisiaca	Musa acuminata x Musa balbisiana	Grand nains	Common banana - paradisiaca cultivars	Musa acuminata x Musa balbisiana	Cavendishes paradisiaca\$Latundan bananas\$Goldfinger banana
A0DQM	Cavendishes acuminata	Musa acuminata Colla	Grand nains	Common banana - acuminata cultivars	Musa acuminata Colla	Cavendishes acuminata\$Dwarf cavendish\$Red banana\$Lakatan banana\$Señorita banana\$Grand nain
A01LE	Plantains			Plantains	Musa acuminata x Musa balbisiana\$Musa x paradisiaca	Cooking banana

In addition to the above changes related to the varieties of Musa spp., the following two codes were identified as redundant and therefore they were dismissed; [A01LD] Dwarf bananas and [A0DQL] Cavendishes - balbisiana cultivars.

3.5. Changes in line with the updated Annex I of the Regulation (EC) No 396/2005

FoodEx2 catalogue was revised in line with the updated Annex I of the Regulation (EC) No 396/2005 related to pesticides³. 19 new terms were identified and added to the catalogue in the MTX (FoodEx2 Matrix), Reporting and Exposure hierarchies as well as in the F27 - Source commodities and F04 - Ingredient facets (see Appendix A for more details). Matrix codes were updated for 50 terms and these changes are presented in Table 4.

Table 4: Updates in the matrix codes of existing FoodEx2 terms in line with the updated Annex I of the Regulation (EC) No 396/2005 related to pesticides

Code	Term extended name	Old matrix code	New matrix code
A00FV	Brussels sprouts	P0242010-000	P0242010-001
A00GT	Sea kale	P0243990	P0243010-007
A00MB	Radish leaves (including radish tops)	P0251080-990	P0243020-008
A00HL	Tree onions	P0220040-002	P0220040-001
A00LZ	Baby leaf indian mustard leaves	P0251080-002	P0251080-003
A01AJ	Galangal roots	P0840020-002	P0213040-005
A01AB	Ginger roots	P0840020-000	P0213040-006
A010L	Greater galangal roots	P0840020-004	P0213040-007
A010M	Lesser galangal roots	P0840020-002	P0213040-008
A015A	Canarium nuts (indicum)	P0120010-004	P0120010-003
A04AD	Pig msm	P1010000	P1011000
A049T	Bovine msm	P1010000	P1012000
A04AM	Sheep msm	P1010000	P1013000
A04AX	Goat msm	P1010000	P1014000
A04BF	Horse msm	P1010000	P1015000
A04DK	Generic poultry msm	P1010000	P1016000
A04DT	Chicken msm	P1010000	P1016000-002
A04EM	Duck msm	P1010000	P1016000-004
A04EX	Goose msm	P1010000	P1016000-005
A04ED	Turkey msm	P1010000	P1016000-016
A04FF	Ratites msm	P1010000	P1016000
A04AA	Pig carcasse	P1010000	P1011000
A049Q	Bovine carcasse	P1010000	P1012000
A04AJ	Sheep carcasse	P1010000	P1013000
A04AS	Goat carcasse	P1010000	P1014000
A04BC	Horse carcasse	P1010000	P1015000
A04BL	Asses-mules-hinnies carcasse	P1010000	P1015000
A04BV	Rabbit carcasse	P1010000	P1017000-016
A04CE	Deer carcasse	P1010000	P1017000
A04DG	Generic poultry carcasse	P1010000	P1016000
A04DQ	Chicken carcasse	P1010000	P1016000-002
A04EJ	Duck carcasse	P1010000	P1016000-003
A04ES	Goose carcasse	P1010000	P1016000-004

³ Official Journal of the European Union, Commission Regulation (EU)2018/62, replacing Annex I to Regulation (EC) No 396/2005 of the European Parliament and of the Council https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:32018R0062&from=EN#ntr12-L_2018018EN.01000302-E0012

Code	Term extended name	Old matrix code	New matrix code
A04EA	Turkey carcase	P1010000	P1016000-016
A04FC	Ratites carcase	P1010000	P1016000
A0CGX	Ginger roots and similar-	P0840020	P0213040
A0CHH	Wasabi roots	P0840020-005	P0213040-010
A0CYC	Galanga roots	P0840020-003	P0213040-005
A0CYD	Fingerrot roots	P0840020-001	P0213040-004
A0DKL	Baby leaf escaroles	P0251080-004	P0251080-002
A0DKM	Baby leaf lettuces	P0251080-003	P0251080-004
A0DRS	Limequats	P0161040-001	P0110040-005
A13GZ	Aegle marmelos (L.) Corrêa (as plant)	-	P0162020-005
A16FL	Chinese water chestnut	P0212990	P0212040-002
A16FZ	Chia seeds	P0401990	P0500020-006
A14LA	Fallopia japonica (Houtt.) Ronse Decr. (as plant)	-	P0270080-001
A000C	African millet grain	P0500040	P0500040-003
A018L	Ajowan fruit	P0810990	P0820030-001
A018T	Caraway fruit	P0820030-000	P0820030
A00RH	Oca	P0213990	P0213050-003

In order to enable correct codification of the pesticides data 48 terms under Pulses (dried legume seeds) [A012R] were transformed to Raw Primary Commodities (RPC), instead of RPC Derivatives, as these are not fresh legumes that were subjected to a drying process, but they are harvested as dry from the plant (Table 5).

Table 5: FoodEx2 codes where the term type was changed

Code	Term extended name	Old term type	New term type
A013A	Lima beans (dry)	d	r
A013B	Rice beans (dry)	d	r
A013C	Black gram (dry seeds)	d	r
A013D	Mung beans (dry)	d	r
A013E	Azuki beans (dry)	d	r
A013F	Tepary bean (dry seeds)	d	r
A013G	Mat bean (dry seeds)	d	r
A013H	Broad beans (dry)	d	r
A013J	Garden peas (dry)	d	r
A013K	Marrowfat pea (dry seeds)	d	r
A013L	Field (maple) pea (dry seeds)	d	r
A013M	Chickpeas (dry)	d	r
A013N	Black eyed peas (dry)	d	r
A013P	Pigeon peas (dry)	d	r
A013Q	Lentils (dry)	d	r
A013R	Bambara groundnut (dry seeds)	d	r
A013S	Lupins (dry)	d	r
A013T	Other pulses (e)	d	r
A013V	Chickling vetches (dry)	d	r
A013X	Vetches (dry)	d	r
A013Y	Jack beans (dry)	d	r

Code	Term extended name	Old term type	New term type
A013Z	Lablab beans (dry)	d	r
A014A	Horse gram (dry seeds)	d	r
A014B	Kersting's groundnut (dry seeds)	d	r
A012R	Pulses (dried legume seeds)	d	r
A012S	Beans (dry) and similar-	d	r
A012T	Borlotti or other common beans (dry)	d	r
A012V	Kidney bean (dry seeds)	d	r
A012X	Navy beans (dry seeds)	d	r
A012Y	Field beans (dry seeds)	d	r
A012Z	Flageolets (dry seeds)	d	r
A04HD	Other pulses	d	r
A0DBV	Andean lupin (dry)	d	r
A0DBX	Yellow lupin (dry)	d	r
A0DBY	Blue lupin (dry)	d	r
A0DBZ	White lupin (dry)	d	r
A0DCA	Lupins (dry) and similar-	d	r
A0DCB	Moringa (dry)	d	r
A0DCC	Asparagus peas (dry)	d	r
A0DCD	Peas (dry) and similar-	d	r
A0DCE	Lentils (dry) and similar-	d	r
A0DCF	Yardlong beans (dry)	d	r
A0DCG	Stink beans (dry)	d	r
A0DCH	Soyabeans for consumption (dry)	d	r
A0DCJ	Runner beans (dry)	d	r
A0DCK	Monantha vetches (dry)	d	r
A0DCL	Guar beans (dry)	d	r
A0DCM	Ervils (dry)	d	r

d: Raw primary commodity derivative; r: Raw primary commodity

The matrix codes present under the implicit attributes of RPC derivatives and simple composite, were removed. As it refers to processed foods, the matrix code remained only for the baby food. The match with the relative RPC present in the Annex I & II of Regulation (EC) No 396/2005 are based on the F27 - source commodity facet.

3.6. Updates in the hierarchy of botanicals of FoodEx2

EFSA's Compendium of botanicals is a database⁴ of botanicals that are reported to contain naturally occurring substances of possible concern for human health when present in food. The compendium is intended to facilitate hazard identification by providing information on composition, toxicity and reported adverse effects for around 2700 plants. First and second versions of the compendium were published in 2009 and 2012. In 2016, EFSA developed a web-searchable version of its Compendium of Botanical with the information for 900 plants made available to the public. 1700 additional plants species have been retrieved from the previous version of the Compendium and have been subject to a systematic literature search and assessment of potential toxicity/adverse effect or evaluation of the presence of substances of potential concern. The taxonomy has been checked through publicly available databases⁵ and the identity of the substances of possible concern for human health has also

⁴ <https://www.efsa.europa.eu/en/data/compendium-botanicals>

⁵ www.theplantlist.org, <https://www.ars-grin.gov/>

been characterised. The collected information is currently being validated and will be added to the database in 2019 once the EFSA FoodEx2 catalogue has been updated accordingly.

Following this assessment, 118 existing terms in the MTX (FoodEx2 Matrix) hierarchy became reportable to the Botanicals hierarchy and 314 new terms (9 new hierarchy terms related to families and 305 extended terms related to different species) were added to both the MTX (FoodEx2 Matrix) and Botanicals hierarchies as live plant organism with the “as plant” extension (see Appendix A for the detailed list of terms).

3.7. FoodEx2 update in accordance with Avian Influenza Surveillance Programmes

EFSA has been assigned the task of collating, validating, analysing and summarising the results from the Avian Influenza surveillance programmes in poultry and wild birds carried out by Member States according to the Commission Decision 2010/367/EU⁶ and guidelines provided by Council Directive 2005/94/EC⁷. In order to enable the data collection FoodEx2 catalogue was populated with 118 new terms under the hierarchy term Bird (as animal) [A057X] in the MTX (FoodEx2 Matrix) hierarchy and in the F01 - Source facet presented in the Appendix A. The selection of these terms was based on the scientific report on the Reporting Avian Influenza surveillance (EFSA, 2018) and the list of wild birds listed in the annual reports available on the Commission's website related to the Avian influenza reporting⁸. Each new term was attributed with the EURINGS code⁹. EURING codes were also attributed to 27 already existing terms, reported as necessary for the Avian Influenza Data Collection. Finally, three codes related to testing swabs were added in the F02 – Part-nature facet.

3.8. Terms that were deprecated, dismissed or set as not reportable

During the maintenance of the FoodEx2 catalogue, 36 terms were deprecated; 28 under the Fish (as animal) [A0BXL] category in the MTX (FoodEx2 Matrix) hierarchy, because they were found as duplicates of other terms and six from the Botanicals hierarchy, because they were identified as synonyms of other existing terms. The existing terms were then updated for the identified synonym. Other two terms were identified as duplicates during the revision of the catalogue according to the latest amendment of the Annex I of the Regulation (EC) No 396/2005. All deprecated terms are presented in Table 6.

Table 6: Terms that were deprecated during the 2016-2018 maintenance

Code	Term extended name
A000C	African millet grain
A0CHD	Black caraway fruit
A04TM	Black bass (freshwater) (as animal)
A053A	Northern bluefin tuna (as animal)
A07ZL	Striped dwarf catfish (as animal)
A086K	Indian mottled eel (as animal)
A087L	Pontic shad (as animal)
A0892	Dolly varden (as animal)
A08HK	Brazilian menhaden (as animal)
A08LT	Eel catfish (as animal)
A0954	Scaled sardines (as animal)
A097D	Bonnethead (as animal)

⁶ Official Journal of the European Union, Commission Decision of 25 June 2010 on the implementation by Member States of surveillance programmes for avian influenza in poultry and wild birds (notified under document C(2010) 4190) (Text with EEA relevance) (2010/367/EU)

⁷ COUNCIL DIRECTIVE 2005/94/EC of 20 December 2005 on Community measures for the control of avian influenza and repealing Directive 92/40/EEC

⁸ https://ec.europa.eu/food/animals/animal-diseases/control-measures/avian-influenza_en

⁹ <https://blx1.bto.org/euringcodes/species.jsp>

Code	Term extended name
A09D4	Norwegian skate- (as animal)
A0A0A	Spinycheek crayfish (as animal)
A0A6S	Atlantic deepsea Lobster (as animal)
A0A7P	Australian pincer Lobster (as animal)
A0ADC	Indian white prawn (as animal)
A0AGA	Southern rough Shrimp (as animal)
A0ANB	Tulip snail (as animal)
A0AQ7	Sunray Venus (as animal)
A0AQ8	Variegated carpet shell (as animal)
A0AQB	Chamber venus (as animal)
A0AXX	Korean mussel (as animal)
A0B06	Sydney cupped oyster (as animal)
A0B0Y	Royal cloak scallop (as animal)
A0B7B	John Dory (as animal)
A0B82	Banded murex (as animal)
A0B8G	Spanish mackerel (as animal)
A0B0N	Delicate scallop (as animal)
A0CKB	Daphnia (as animal)
A13QN	Arum montanum Roxb. (as plant)
A13XL	Brunfelsia uniflora (Pohl) D.Don (as plant)
A141Q	Chamaecrista nomame (Sieber) H.Ohashi (as plant)
A144Q	Clinopodium nepeta (L.) Kuntze subsp. nepeta (as plant)
A14ZR	Ledum palustre var. palustre (as plant)
A15QX	Smilax tonduzii Apt (as plant)

Finally, 38 terms were identified as redundant in specific hierarchies and facets and, therefore, marked as dismissed or not reportable. Nine of these were dismissed because they were inconsistent with the logic of the system or were identified as not at all –or very rarely- marketed as such. These are listed in Table 7. The remaining 29 are hierarchy terms (blue pyramids in the browser tool) of the Reporting hierarchy. From 2019 onwards, chemical occurrence data will be submitted to EFSA according to the Reporting hierarchy of FoodEx2, therefore, these terms were set as not reportable in the specific hierarchy in order to prevent data providers from using them and consequently respect the rule set up in the EFSA guidance (EFSA, 2015) where it is highlighted that hierarchy terms should never be used.

Table 7: Terms that were dismissed during the 2016-2018 maintenance

Code	Term extended name
A003E	Rice grain, parboiled
A011P	Potato boiled
A011R	Potato baked
A03LJ	Jasmine infusion
A03LM	Rose infusion
A03LQ	Strawberry leaves infusion
A03LV	Ginkgo infusion
A03LX	Valerian root infusion
A03LY	Ginseng root infusion

3.9. Revision of applicability or position of existing terms in specific hierarchies and/or facets

In order to adapt the catalogue according to the requestors' needs some of existing terms were added to the particular hierarchies and facets. For example, as described in section 3.4 of this report, some of the existing terms were added to the Botanicals hierarchy. In line with changes described in section 3.2 of this report, related to the pulses, some of the existing terms were added to the section Pulses (dried legume seeds) in the facet F27 - Source-commodities.

Due to revision of the position for existing terms, parent code was changed for some of them in one or more hierarchies and/or facets. These minor changes are not presented in this report, but they were implemented, therefore they are included in the FoodEx2 database and appear in the Appendix file 'MTX_10.0.xlsx' spreadsheet 'term' that contains the complete FoodEx2 catalogue after the maintenance 2016-218.

3.10. Additional activities on the FoodEx2 database

3.10.1. Mapping from FoodEx2 to FoodEx

The mapping between FoodEx2 codes and FoodEx codes was carried out in order to convert new occurrence and consumption data submitted to EFSA in FoodEx2 back to FoodEx. This conversion is necessary during the transitional period between the two systems, when data are still analysed based on their FoodEx codes. The FoodEx2 codes considered in the mapping are mainly the reportable terms of the Exposure and Reporting hierarchies. Details about mapping are published in the Technical Report: Mapping FoodEx2 to FoodEx1 published in 2016 (EFSA, 2016b). The resulting mapping was integrated, as implicit attribute, in the field 'foodexOldCode' of FoodEx2 catalogue.

3.10.2. Revision of the implicit facets

A major revision of the FoodEx2 catalogue was performed related to the implicit facets. Implicit facets are description associated with a term, without the need to insert them at the time of coding. The previous version of FoodEx2 only included implicit facets in the area of source and part-nature facets and still these were not attributed to all terms present in the catalogue. Therefore, resources have been allocated to check and update all implicit facets in the catalogue and in the Data Collection Framework (DCF) platform. This update was done in several steps. Firstly, the existing implicit facets and their correctness were checked considering the possible uses of the codes based on the rules established for coding with FoodEx2. Missing implicit facets were identified and added to the catalogue and its updated version was imported in DCF. The consistency of the pre-defined combination of terms and implicit facets were tested successfully for the MTX (FoodEx2 Matrix) and Zoonoses hierarchies.

Changes applied in the area of implicit facets are listed in the Table 8. In this table the changes are grouped per facet (first column) and applicability (second column). The third column describes the number of occasions the same facet can be attributed for one single term.

As a rule, each term automatically adopted the implicit facet F01 – Source from its parent, when a more specific was not already selected. In addition, the implicit facet F01 – Source was attributed to all terms defined as RPC. The facet F02 – Part-nature was also attributed to all terms of the catalogue. The facet F27 – Source-commodities was attributed to all RPC derivatives but only when it was clearly implicit (e.g. Dried apricots – F27: Apricots) and to RPC where this implicit facet is the term itself (e.g. Apples – F27: Apples).

Terms with implicit process of preparation are attributed with the facet F28 – Process (e.g. Sauerkraut – F28: Fermentation). The facet F04 – Ingredient was attributed to terms defined as composites and when the characterising ingredient was evident from the term extended name (e.g. Fish and rice meal – F04: Fish (meat) and Rice grain). Similar was the case for the F06 – Surrounding medium facet (e.g. Fruit in alcohol or spirits – F06: In alcohol-spirit-liqueur), facet F03 – Physical-state (e.g. oat flour - F03: powder facet F09 – Fortification-agent (e.g. Salt, iodized – F09: Iodine) and facet F10 – Qualitative-info (e.g. Cola beverages, decaffeinated – F10: Decaffeinating). The facet F20 Part-consumed-analysed was attributed only to the terms where the consumed part of the food is indicated

in the term extended name as well as facet F21 Production-method. Terms related to the food items for targeted consumers were attributed with the facet F23 – Targeted-consumer such as Infant and follow-on formulae [A03PY] – F23: Infant or toddler food. The facet F24 Intended-use was attributed to the terms where physical state of the food is indicated in the term extended name and where the intended use must be specified (e.g. Pasta for children (dry, to be cooked) – F24: Intended to be eaten cooked). Terms related to the animals with implicit purpose of raising, reproductive level, age class or gender are attributed with one or more of the following facets: F29 - Purpose-of-raising, F30 - Reproductive-level, F31- Animal-age-class and F32 - Gender.

Table 8: Defined occasions when implicit facets were added to the term

Name	Implicit facet*	Facet schema
Source (F01)	For RPC ^(a)	One
Part-nature (F02)	Always ^(b)	One
Physical-state (F03)	Only in case of e.g. yogurt drink: liquid	One
Ingredient (F04)	In composite, just when clearly implicit ^(c) (characterising ingredient) (in derivatives only as minor ingredient)	Many
Surrounding-medium (F06)	Only in case of e.g. Fruit in alcohol or spirits	Many
Fat-content (F07)	No	One
Sweetening-agent (F08)	No	Many
Fortification-agent (F09)	Only in case of e.g. Salt, iodized	Many
Qualitative-info (F10)	Only in case of e.g. light, decaffeinated, whole meal, gluten free products	Many
Alcohol-content (F11)	No	One
Dough-Mass (F12)	Just the ones always implicit for the specific term	Many
Extent-of-cooking (F17)	No	Many
Packaging-format (F18)	No	One
Packaging-material (F19)	No	Many
Part-consumed-analyzed (F20)	Just for A01QC Candied fruit, orange peel A0F1J Edible casings A0F1H Cheese rind	Many
Production-method (F21)	Just for e.g. A01TF Wild goose fresh meat A0CTX Wild terrestrial vertebrate animals	Many
Preparation-production-place (F22)	No	One
Target-consumer (F23)	Just for e.g. infant/toddler, weight reduction products, food for sport people	Many
Intended-use (F24)	Only in case of e.g. Pasta for children (dry, to be cooked) or Pasta-based dishes, uncooked	One
Risky-Ingredient (F25)	No	Many
Generic-term (F26)	No	One
Source-commodities (F27)	In derivatives, just when clearly implicit and in RPC (themselves)	Many
Process (F28)	Just the ones always implicit for the specific term ^(d)	Many
Purpose-of-raising (F29)	In animals raised for different purposes such as Calves that could be raised for milk production e.g. Dairy calves or meat production e.g. Veal calves.	Many
Reproductive-level (F30)	In animals where, reproductive level is indicated such as parent, grandparent etc. e.g. Elite Gallus gallus breeding flock	One
Animal-age-class (F31)	In animals where age class is indicated such as e.g. Dairy calves (under 1 year)	Many
Gender (F32)	In animals where gender is determined by default such as e.g. Young bulls or Heifers	One
Legislative-classes (F33)	No	Many

(a): exception for e.g. A0C9E Calves (under 1 year) and similar where the F01 was added as implicit facet even if the type of term is set as natural sources, and for A038L Processed fat (blubber) of marine mammals where the F01 was added to specify the F27 implicit facet present

- (b): exception for e.g. A0CPA Chromista (live organisms) where the F02 missing
 (c): exception for e.g. some RPC derivatives where the F04 fact was added as minor ingredient
 (d): A0C02 Oil production: used also for animal fat

3.10.3. Revision and enrichment of the scope notes

A complete revision of the scope notes was implemented during the 2016-2018 maintenance of the FoodEx2 catalogue with the purpose of improving the information they provide and including external links that can help the user identify the food/feed item intended to be reported. This activity impacted on the corresponding field ('Scopenotes') of FoodEx2. The links added are redirecting the user to the e.wikipedia.org and google.co.uk websites for general terms as well as additional websites for more specific terms, e.g. fishbase.org, marinespecies.org, etc. These changes are not presented in detail in this report, but they were implemented, therefore they are included in the FoodEx2 database and appear in the Appendix file 'MTX_10.0.xlsx' spreadsheet 'term' that contains the complete FoodEx2 catalogue.

3.10.4. Mapping of FoodEx2 facet descriptors with LanguaL codes

LanguaL¹⁰ stands for "Lingua Alimentaria" or "language of food". It is an automated method for describing, capturing and retrieving data about food and many food composition databases worldwide have been indexed according to LanguaL. The most current version of LanguaL, version 2017, was published in April 2017. In this last version LanguaL has been updated extensively - especially in the area of facets product type and food source. FoodEx2 was also included in this last version.

In the course of 2018, EFSA became partner in a new GODAN Working Group (WG) on "Publication and Alignment of Authoritative Vocabularies for Food"¹¹. This WG is bringing together the key domain authorities and other partners who can provide the platforms and methodologies to work on and publish vocabularies. Its final objective is to provide a set of semantically interlinked URIs for key food product concepts as an infrastructural component that will facilitate the development of both software for actors in the food value chain and added-value integrated services that need to track food products. Under the umbrella of this WG, an update and further mapping of the FoodEx2 facet terms with corresponding LanguaL codes implemented in the first release of the catalogue took place. The mapping concerned 13 facets. The associated LanguaL term was added as implicit attribute to the FoodEx2 term and therefore, a search based on the LanguaL code is now enabled. The number of associated codes is presented in Table 9.

Table 9: Update of the FoodEx2 mapping with LanguaL codes

FoodEx2 facet	LanguaL codes added (n)	LanguaL codes changed (n)	LanguaL codes removed (n)
Source (F01)	2192	88	15
Process (F28)	21	5	1
Surrounding-medium (F06)	2	1	
Qualitative-info (F10)	38	5	1
Production-method (F21)	1		
Packaging-material (F19)	11		
Physical-state (F03)	10	2	
Part-consumed-analyzed (F20)	12	3	
Target-consumer (F23)	2	1	
Total	2289	105	17

3.10.5. Development of the FeedAddExpo hierarchy

The FeedAddExpo hierarchy was developed to be used for the Feed Additives Consumer Exposure (FACE) calculator. This tool allows the estimation of dietary exposure to residues of feed additives

¹⁰ <http://www.languaL.org/Default.asp>

¹¹ <https://www.godan.info/news/new-godan-wg-food-classifications>

(and their metabolites) present in food of animal origin. The FeedAddExpo hierarchy facilitates the input of chemical occurrence data into the calculator and is compliant with the grouping of residue data recommended by the Guidance on the assessment of the safety of feed additives for the consumer (EFSA, 2017). This new hierarchy is neither reportable nor available for usage in any other data collection.

4. Conclusions and recommendations

Since its first release, the FoodEx2 food classification and description system of EFSA has been revised two times. Even if a good level of comprehensiveness and stability after those revisions was achieved, continuous changes in the food market and the experience gained by users and stakeholders imply the evolution of the system. Therefore, a maintenance is recommended in order to meet new needs required in different food safety domains. As a result, between 2016 and 2018 the FoodEx2 catalogue was updated following requests from users and needs of various data collections.

Based on suggestions provided by users and stakeholders and requests for addition of missing terms, the 2016-2018 FoodEx2 maintenance was carried out:

- New terms (680) were added in different hierarchies and facets
- The names of some existing terms were changed to better specify the scope of the term or to expand its meaning
- Alias – included terms were added to existing terms to clarify their intended use
- A major revision was undertaken in the botanicals hierarchy, and within sections related to the insects and arachnids, birds, pulses and varieties of *Musa* spp.
- Revision of Matrix codes was done according to the latest update of Annex I of the Regulation (EC) No 396/2005
- 36 duplicate terms were deprecated
- The use of 38 terms was changed into dismissed or not reportable in certain hierarchies and facets due to the inconsistencies with the logic of the system
- The applicability of some terms was extended to further hierarchies and/or facets
- The position of some terms in the hierarchy tree was changed since a better parent-child term relationship was found

While carrying out this maintenance, some improvements on the scope of the terms (i.e. the description of the food group recorded in the field 'Scope Notes') and some amendments on the implicit facets were also carried out. Furthermore, the mapping between FoodEx2, and LanguaL Thesauri was revised and updated.

The present report documents the update of FoodEx2 during the years 2016-2018. Therefore, it should be read in conjunction with the technical report 'The food classification and description system FoodEx2 (revision 2)' (EFSA, 2015) where the logic and the structure of the system and the rules for the correct use of FoodEx2 are described. These are essential elements to perform a correct coding with a reasonable level of detail.

The complete content of the FoodEx2 system is available as an Appendix to this document as exported spreadsheet MTX_10.0.xlsx¹².

The experience gained in these years confirms the need of an annual maintenance and updating cycle for FoodEx2 in order to take into account the evolution of the food market and the new needs in classifying food in the different food safety domains.

¹² The MTX_10_0.ecf is available at <https://github.com/openefsa/catalogue-browser/wiki/Use-the-tool-without-DCF-account>

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Abbreviations

BSE	Bovine Spongiform Encephalopathy
DATA Unit	Evidence Management Unit
DCF	Data Collection Framework
EFSA	European Food Safety Authority
EC	European Commission
EU	European Union
FAO	Food and Agriculture Organization of the United Nations
FACE	Feed Additives Consumer Exposure
INFOODS	International network of Food Data Systems
RPC	Raw primary commodity
TSE	Transmissible Spongiform Encephalopathy
WG	Working Group

Appendix A – New terms added during the 2016-2018 maintenance

Appendix B – FoodEx2 catalogue MTX_10.0